# Youchul Jeong

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## Education

University of the Aegean, Department of Marine Science — Mytilene, Greece
Ph.D. Candidate in Marine Science — *Oct 2025 - Present*
- Supervisor: Assoc. Prof. Konstantinos Topouzelis
- Marine Remote Sensing Group (MRSG) — *May 2025 - Present*

Pusan National University, Department of Earth Environmental System — Busan, South Korea
**Master of Science in Oceanography** — ***Feb 2024***
- MSc Thesis "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." (Supervisor: Prof. Young-Heon Jo)
- Oral presentation award, Ocean & Atmosphere Science Interdisciplinary Studies Conference
- BrainKorea21 Scholarship (Earth Environmental Systems Education Research Group)

Pusan National University, Department of Oceanography — Busan, South Korea
**Bachelor of Science in Oceanography** — ***Feb 2022***
- BSc graduation paper "A Study on Image-Based Marine Litter Detection Techniques." (Supervisor: Prof. Young-Heon Jo)
- Center of Remote Sensing Lab — *Jan 2021 - Apr 2025*
- Academic Excellence Scholarship from the Department of Oceanography

## Research Appointments

**Scientific Researcher** — *May 2025 - Oct 2025*
Department of Marine Science, University of the Aegean — Mytilene, Greece
- Floating marine debris research work using drone and satellite imagery in the Marine Remote Sensing Group (MRSG)

**Research Assistant** — *Mar 2024 - Apr 2025*
Marine Research Institute, Pusan National University — Busan, South Korea
- Conducted drone fieldwork and developing a deep learning model to detect marine debris in the ocean.

**Research Assistant** — *Sep 2022 - Dec 2022, Sep 2023 - Feb 2024*
Marine Research Institute, Pusan National University — Busan, South Korea
- Developed innovative methodologies including drones, AI, and multi-spectral sensors to contribute to published articles on remote sensing.

## Research Experience

**Plastic Litter Project 2025 (PLP2025)** — *May 2025*
- Constructed floating plastic-litter targets using HDPE tarps and wooden planks in 1 $m^2$, 2 $m^2$, and 3 $m^2$ configurations for the PLP2025 field experiment.

- Assisted with field deployment, monitoring, repair, and retrieval of six target units in the Gulf of Gera, Lesvos.

**Drone Analysis of Beach Safety and Beachgoer Behavior** *Nov 2023, Feb - Aug 2024*
- Orchestrated drone flights and implemented the YOLOv8 algorithm to observe beach safety.
- Conducted surveys of beachgoer population to understand motivations and improve beach conditions.

**Classification and Monitoring of Floating Marine Macro Litter (MSc Thesis)** *Jun 2022 - Dec 2023*
- Utilized drones with multispectral sensors to create a custom FMML dataset.
- Implemented deep learning models for efficient litter classification and continuous monitoring.
- Incorporated drone atmospheric correction using DROACOR in collaboration with Dr. Daniel Schläpfer (ReSe Application LLC).
- Worked with advisors to promote conceptual framework and fieldwork at international conferences.

**Floating Marine Debris and Physical Ocean Phenomena in S-ORS** *April 2021 - Dec 2023*
- Applied drone technology and Applied Drone Automatic Flight Advisory in open ocean conditions.
- Developed a CNN model for detecting marine debris for Socheongcho Ocean Research Station.

**Drone Survey of Spotted seal Population** *May 2022 - Nov 2023*
- Conducted drone surveys to assess spotted seal population on Baekryeong Island and gather high-resolution imagery, contributing crucial data to understanding spotted seal dynamics.

**Floating Marine Litter, Korea Institute of Ocean Science & Technology** *May 2023 - Oct 2023*
- Worked with Dr. Won Joon Shim (KIOST) to conduct on-site surveys of microplastics and floating marine litter.
- Compared and analyzed results of drone and multispectral sensor surveys.

**Coastal Topography and Seabed Survey** *Jul 2023 - Aug 2023*
- Conducted RTK-GPS and drone surveys to analyze coastal sand loss and seabed depth change.

**UAV Team Leader, Korean Air Force** *May 2016 - May 2018*
- Assembled military training UAVs, managed GPS element procurement, and led team activities.

## Teaching Experience

**Interpretation of marine data [Course No. MS3200990]** *1st semester 2023*
- Led fieldwork for 40 undergraduate students to acquire Sargassum multi-spectral images.
- Taught data processing using MATLAB as part of the project curriculum.

**Coastal Practices Survey [Course No. MS3200295]** *1st semester 2022, 2023*
- Delivered theoretical lectures on YOLOv5, covering installation process and algorithms.

## Peer-Reviewed Publications

**Youchul Jeong**, Jisun Shin, Jong-Seok Lee, Ji-Yeon Baek, Daniel Schläpfer, Sin-Young Kim, Jin-Yong Jeong, Young-Heon Jo. "A Study on Monitoring Floating Marine Macro-Litter using a Multi-Spectral Sensor and Classification based on Deep Learning." Remote Sensing, 2024.

Sin-Young Kim, Jong-Seok Lee, **Youchul Jeong**, Young-Heon Jo. "Decomposition of Submesoscale Ocean Wave and Current Derived from UAV-Based Observation." Remote Sensing, 2024.

## Conference Presentations

**Youchul Jeong** et al., "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." Poster presentation at 11th Asian Workshop on Ocean Color, Nagoya, Japan, 2023.

**Youchul Jeong** et al., "A Study on Monitoring Floating Marine Macro Litter using Multi-spectral Sensor and Classification based on Deep Learning." Oral presentation at Korea Ocean Research Conference, Seoul, South Korea, 2023.

**Youchul Jeong** et al., "A Study on Monitoring Floating Marine Macro Litter using a Multi-spectral Sensor and Classification based on Deep Learning." Oral presentation at Ocean & Atmosphere Science Interdisciplinary Studies Conference, Busan, South Korea, 2023.

**Youchul Jeong** et al., "A Study on Classification and Monitoring of Floating Marine Macro Litter using Multispectral Imagery in UAV and Deep Learning." Poster presentation at European Geoscience Union General Assembly, Vienna, Austria, 2023.

**Youchul Jeong** et al., "A Study on Classification and Monitoring of Floating Marine Macro Litter using Multispectral Imagery in UAV and Deep Learning." Oral presentation at 10th Asian Workshop on Ocean Color, Indonesia, 2022.

## Technical Skills

- Programming and analysis: MATLAB, Python
- Remote sensing software: ENVI 5.2, DROACOR, ACOLITE
- Satellite imagery: PRISMA, EnMAP
- UAV operations: 100+ research flights; South Korean UAV license for 25-150 kg aircraft
- Languages: Korean (native), English (proficient), Greek (beginner)